bare_jrnl_new_sample4-37.pdf Introduction 修改方案

动态机器人减少 + 与 H-LTLf / 子公式交集类方法的差异化表述

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结论先行。 Introduction 的修改方向不应把“机器人动态减少”包装成新的理论创新，而应作为实际执行中的可用机器人集合变化来简短引入；真正需要强化的是：你的方法不是通过要求用户重新建模成 H-LTLf、层次结构或多个子公式交集来降低复杂度，而是在**给定全局时序逻辑公式**的自动机构造过程中嵌入机器人、工作空间和中间规划信息，从**方法层**提前剪枝并快速得到初始解，再继续优化。因此，建议把 gap 改成“现有方法要么依赖特殊建模结构/人工分解，要么关注动态任务重构；而对于一般全局 LTL-MRTA，如何在不先完整生成自动机和规划结构的情况下快速得到可执行且可继续优化的方案，仍然不足”。

1 现有 Introduction 的主要问题

从严格审稿人的角度看，当前 Introduction 的主线已经能说明 LTL-MRTA 的计算复杂性和你方法的基本思路：你的稿件明确写到，现有方法通常先构造完整 NBA 再规划，而你的方法在 LTL2BA 的 $\varphi \to T_\varphi \to V_\varphi \to G_\varphi \to B_\varphi$ 管线中引入机器人团队、工作空间和中间规划信息，并在 GBA/NBA 构造阶段进行规划相关剪枝，随后通过 planning graph 和 warm-start branch-and-bound 继续优化。这个主线是合理的，但目前存在三个明显缺口。

1.1 问题一：动态执行情形没有进入问题动机

当前 Introduction 几乎仍按静态任务和静态机器人团队展开。你现在的方法已经包含“机器人动态减少”的处理，如果 Introduction 完全不提，后文出现该能力时会显得突兀；但如果把它写成 contribution，又容易被审稿人追问理论定义、完备性、最优性和实验覆盖。因此应采用中间策略：

- 在动机段中用 2–3 句说明真实执行中机器人数量/可用性可能下降，例如故障、低电量、通信退出、任务执行中能力不可用等。
- 把它定义成 available robot set 的变化，而不是新的 temporal-logic 任务语义。
- 在贡献前的方案概述中只写“the framework can be re-invoked or updated with the currently available robot set”，不要写成“we solve dynamic MRTA”或“guarantee resilience to robot failures”。

1.2 问题二：与 H-LTLf 和子公式交集类方法的差异没有讲清楚

当前相关工作中仅用一句话提到 H-LTLf，这不足以回应审稿人可能提出的质疑：“既然 H-LTLf 或 Poset-prod 已经能处理长公式，为什么还需要你的方法？”这个问题必须正面回答，但回答方向不能是“我们的表达能力更强”或“所有公式都能处理得更好”，而应是：

- H-LTLf 和 Poset-prod 主要从**建模结构/公式表示**入手，降低长公式带来的复杂度；
- 这些方法通常需要任务公式具有或被转换成层次结构、多个相对独立的子公式、或适合 R-poset 乘积的合取结构；

- 对于紧耦合协作任务、跨子公式共享机器人资源、强同步约束、交错依赖、或难以自然拆成独立子公式的任务，人工构造层次/交集结构并不总是直接；
- 你的方法从**算法流程**入手，不要求用户预先给出特定层次结构或子公式交集形式，而是在自动机构造过程中利用机器人可行性和规划信息剪枝。

1.3 问题三：gap 需要从“快”提升为“快且不依赖特殊公式结构”

当前 gap 主要是“existing methods struggle to simultaneously achieve rapid first-solution generation and optimization”。这句话是对的，但还不够尖锐。建议把 gap 改成三层：

1. 传统 product automaton / MILP / search 方法：精确或通用，但完整自动机和组合搜索代价高；
2. H-LTLf、Poset-prod、partial-order 等：能缓解长公式，但依赖任务结构提取、层次化建模或子公式合取分解；
3. reactive/dynamic 方法：能处理任务增加/删除或语义触发，但重点是任务逻辑更新/重规划，不是“在全局 LTL-MRTA 的自动机构造过程中快速生成可优化初始方案”。

建议用一句话统领 gap：**“What remains insufficiently addressed is a planning method that can start from a standard global temporal-logic task, avoid relying on a manually imposed hierarchical or conjunctive decomposition, and still return an executable allocation early while preserving a structure for subsequent optimization.”**

2 动态内容应该如何补充

2.1 应该补充什么

建议把动态内容限定为两类：

1. **任务需求动态变化**。任务增加/删除可作为相关工作的背景出现，用来说明真实系统中 temporal-logic 任务并不总是一次性离线给定。上传的 reactive MRTA 论文明确把 sudden addition/removal of tasks 作为动态需求，并在检测到语义 landmark 后触发 replanning；Poset-prod 论文也强调任务公式会 online and continually released/canceled。
2. **机器人可用性动态减少**。这是你要补的重点，但要写得轻。可以说机器人故障、能量不足、能力损失或退出会减少可用机器人集合，使原先可行的协作原子命题变得不可行。上传的 reactive MRTA 论文在 Remark 3 中也采用了类似工程处理：当资源接近耗尽时可将机器人视为 faulty 并从后续分配中排除。

2.2 不应怎么写

不要在 Introduction 中写成：

- "This paper solves dynamic LTL-MRTA with robot failures."
- "The proposed method guarantees resilience against robot loss."
- "Robot removal is a main contribution."

这些写法会把审稿人的注意力引向动态系统理论保证，并要求你补充机器人故障模型、故障发生时间、当前任务是否中断、已执行 trace 如何映射到新 NBA/plan graph、最优性如何保持等问题。

2.3 建议插入位置

建议在当前 Introduction 第二段，也就是解释 LTL-MRTA 难点和"decisions may need to be updated online"之后，加入一小段。该段承上启下：既说明实际动态性，又不改变论文主贡献。

可直接使用的英文段落 1：动态执行背景

In real deployments, the execution context is not always fixed after the initial plan is computed. Newly observed events may add or remove task requirements, and the available robot team may decrease due to failures, low energy, communication loss, or temporary capability loss. Such changes modify the set of executable collaborative propositions and may invalidate assignments that were feasible under the initial team. In this paper, we primarily focus on efficient planning for a given global LTL-MRTA instance, while allowing the planning procedure to be re-invoked with the currently available robot set when robot availability decreases during execution. This practical update is used to support execution robustness, rather than being treated as a separate theoretical contribution.

如果担心 "communication loss" 与你的模型不完全一致，可以删掉，只保留 "failures, low energy, or temporary capability loss"。如果你的方法只是"机器人减少后重新规划剩余任务"，应把 "re-invoked with the currently available robot set" 改为 "applied to the remaining task with the updated available robot set"。

3 与 H-LTLf 和子公式交集/Poset-prod 的差异化表述

3.1 核心定位：它们是建模/分解路线，你是方法/自动机构造路线

H-LTLf 的优势在于把长而复杂的 sc-LTL/LTLf 任务组织成层次结构，从而提升可读性、解释性和搜索效率；Poset-prod 的优势在于当任务可写为许多 sc-LTL 子公式的合取时，分别计算 R-poset 并在线做乘积，从而避免完整合取公式直接转 NBA 的爆炸。它们都很强，但其优势建立在**任务结构被显式组织**这一前提上。

你的优势应提炼为三点：

1. **不要求用户预先把公式写成特定层次或多个子公式交集**。你的输入仍是全局 LTL 任务及协作原子命题，不需要人为决定哪些任务属于同一层、哪些子公式可独立求解、哪些子公式可以安全合取。

2. **保持全局自动机语义下的耦合关系**。对于多个协作子任务之间存在同步、互斥、共享机器人、跨阶段等待、或者前后依赖的情形，强行拆成多个子公式可能降低灵活性或引入额外建模负担；你的方法在 GBA/NBA 构造中直接根据可执行性和规划代价剪枝。

3. **目标不是只得到一个可行解，而是快速初始解 + 后续优化**。 Poset-prod 侧重在线可行规划和多子公式组合，H-LTLf 侧重层次表达与搜索；你的框架强调 planning graph 给出 warm start，再通过 branch-and-bound 改善 makespan。

3.2 建议放入 Related Work 的英文段落

可直接使用的英文段落 2：H-LTLf 与 Poset-prod 类方法

Another line of work reduces the burden of long temporal specifications by changing or exploiting the structure of the task formula. Hierarchical temporal-logic specifications, such as H-LTLf, organize complex specifications into a hierarchy, which improves interpretability and allows the planner to search over loosely connected subspaces. Poset-based methods further exploit the case where an sc-LTL mission is given as a conjunction of many subformulas; each subformula is converted into an R-poset and the products of these posets are computed online to accommodate newly released tasks. These approaches are effective when a suitable hierarchical or conjunctive decomposition is available. However, obtaining such a representation may itself require manual modeling effort or an additional decomposition procedure, and not all collaborative temporal tasks can be cleanly separated into weakly coupled subformulas without losing useful allocation flexibility.

可直接使用的英文段落 3：你的方法与它们的差异

Different from the above representation-level approaches, this work addresses the scalability issue from the automaton-generation and planning side. Starting from a global temporal-logic specification with collaborative action propositions whose robot assignments are not fixed a priori, we inject robot-team, workspace, and partial-planning information into the GBA/NBA construction process. Therefore, automaton branches that are logically valid but infeasible or unpromising for the current heterogeneous team can be pruned before the complete automaton and planning structure are fully materialized. This makes the proposed framework complementary to hierarchical and formula-decomposition methods: it does not require the user to impose a specific hierarchy or intersection form, while still producing an executable initial plan early and preserving a planning graph for subsequent optimization.

这里最关键的是 “complementary”。不要写 “superior to H-LTLf” 或 “more general than Poset-prod”。更稳妥的表述是：H-LTLf/Poset-prod 在合适建模结构下很有效；你的方法解决的是不想或不能预先结构化公式时，如何在自动机构造中提前使用规划信息。

4 建议的 Introduction 结构

位置	现有功能	修改建议
第 1 段	MRS 应用 + temporal constraints	保留。可把应用从 warehouse 扩展到 maintenance / inspection / logistics，但不要过长。
第 2 段	LTL-MRTA 定义 + 计算难点	保留你的例子；在段尾加入“执行中任务和机器人集合可能变化”的 2–3 句。
Related Work 1	bottom-up vs top-down	可略微压缩 bottom-up，因为本文重点是 top-down。
Related Work 2	product automaton / MILP / sampling	保留，但建议把“显式指定机器人导致公式指数变长”的问题说得更清楚。
Related Work 3	decomposition / STAP / H-LTLf / partial-order	重写。把 H-LTLf 和 Poset-prod 放在“representation-level / formula-structure-based methods”中，并说明其依赖层次/合取结构。
Related Work 4	dynamic/reactive methods	新增短段。说明已有方法处理任务增加/删除、语义触发和重规划；你的机器人减少只是可用团队更新，不作为主要创新。
Gap 段	现有方法难以同时快速初始解和优化	改成“无需预设层次/交集结构的全局 LTL-MRTA 中，如何边生成自动机边规划，并快速得到可优化初始解”。
方法概述	proposed framework	保留，但加入“can be re-invoked with updated available robot set”一句，不进 contribution。
Contributions	三条贡献	保持三条理论/算法贡献，不新增动态机器人减少贡献。

5 建议重写的 Gap 段

可直接使用的英文段落 4：最终 gap

Overall, existing top-down LTL-MRTA methods have improved scalability from different angles, including product-automaton search, MILP formulations, sampling-based synthesis, task decomposition, hierarchical specifications, and poset-based products. Nevertheless, a common difficulty remains: before allocation can be effectively optimized, many methods either need to construct a large automaton/planning structure or require the task formula to be manually organized into a hierarchy or a set of weakly coupled conjunctive subformulas. For collaborative heterogeneous tasks with unassigned robots and tightly coupled temporal dependencies, such preprocessing can be expensive or restrictive. Therefore, it remains important to develop a planning method that starts from a global temporal-logic specification, obtains an executable allocation early without fully materializing all automaton branches, and still preserves enough structure for subsequent solution improvement.

这个 gap 可以替换当前 “Overall, while existing top-down methods ...” 那一段。它比当前版本更有说服力，因为它直接回应了 H-LTLf 和 Poset-prod 可能带来的审稿质疑。

6 建议重写的方法概述段

可直接使用的英文段落 5：方法概述

To address this problem, we propose a planning-aware automaton-generation framework for LTL-MRTA. Instead of constructing the complete NBA first and then solving the allocation problem on top of it, the proposed method incorporates robot-team feasibility, workspace travel information, and partial planning results into the LTL-to-automaton translation process. During GBA construction, infeasible or decomposition-inducing transitions are pruned before they are propagated to the NBA stage. During NBA construction, the NBA and a shared planning graph are generated synchronously, so that an accepting prefix-suffix structure can yield an executable initial plan as soon as it becomes available. The resulting plan graph then provides a warm start for a branch-and-bound assignment optimizer, which continues to improve the makespan. When the available robot set decreases during execution, the same planning procedure can be applied to the remaining task with the updated team, serving as a practical plan-repair mechanism.

最后一句可以保留在方法概述段，但不要在 contribution 里单独列出来。如果后文没有实际实现 “remaining task” 而只是重新从头规划，请把最后一句改成：“the same planning procedure can be re-invoked with the updated team.”

7 Contribution 应该如何改

当前 contributions 基本合理，但建议更精炼，并避免把 dynamic robot decrease 写成 contribution。推荐三条如下。

可直接使用的英文 contributions

The main contributions are summarized as follows.

1) We develop a planning-aware automaton-generation framework for LTL-MRTA, in which robot-team, workspace, and partial-planning information is introduced during the LTL-to-automaton translation process rather than only after the complete automaton has been constructed.

2) We propose planning-oriented pruning and on-the-fly plan-graph construction. Infeasible and decomposition-inducing transitions are removed at the GBA level, while the NBA and a shared planning graph are constructed synchronously so that an executable initial solution can be obtained early.

3) We design a warm-start branch-and-bound optimization procedure on the generated plan graph to improve the makespan after the first feasible solution is found. This yields an anytime-style workflow that combines fast initial planning with continued solution improvement.

可以在 contributions 后加一句非贡献性说明: “The implementation also supports updating the available robot set during execution and re-planning the remaining task, which is used in the experiments to handle robot removal.” 但不建议作为第 4 条 contribution。

8 对动态机器人减少的更精确表述

如果你想在 Introduction 中把“机器人动态减少”讲得更自然，可以采用以下四个层次：

1. **动机层：** “robot availability may decrease during execution”。
2. **问题层：**“the set of robots that can satisfy a collaborative action proposition changes over time”。
3. **方法层：** “the planner updates the available robot set and repairs/replans the remaining execution”。
4. **边界层：**“this is a practical execution mechanism, not the main theoretical contribution”。

可直接使用的短句组合

Robot availability may also change during execution. For example, a robot may leave the team due to failure, low energy, or temporary loss of a required capability. Since collaborative action propositions require a specified number of robots of certain types, such team-size reductions may invalidate previously feasible assignments. In the implementation, this situation is handled by updating the available robot set and re-planning the remaining task; this mechanism is used to improve practical execution robustness but is not the main theoretical focus of this paper.

9 需要避免的过度表述

1. 不要说 “not all formulas can be converted into H-LTLf”。更稳妥的是：“not all missions admit a natural or useful hierarchical decomposition without additional modeling effort”。
2. 不要说 “subformula-intersection methods cannot handle coupled tasks”。更稳妥的是：“their efficiency relies on a suitable conjunctive decomposition; tightly coupled tasks may reduce the benefit or require nontrivial modeling”。
3. 不要说 “our method handles arbitrary LTL better”。更稳妥的是：“our method is less dependent on a prescribed formula structure and exploits planning information during automaton generation”。
4. 不要把机器人减少写成 resilience guarantee，除非你在理论部分给出 failure model、trigger condition、remaining-trace satisfaction 和 completeness proof。

10 建议同步修改的论文其他位置

虽然本次重点是 Introduction，但如果 Introduction 增加机器人动态减少，后文至少要有最小配套，否则审稿人会觉得“只在引言里提了但方法没定义”。建议同步做三处轻量修改：

1. **Problem Description**：增加 active robot set，例如 $R^{\text{act}} \subseteq R$，并说明机器人减少时 $R^{\text{act}} \leftarrow R^{\text{act}} \setminus R^{-}$。
2. **Algorithm / Implementation**：增加一句：当检测到机器人不可用时，删除未来分配中涉及该机器人的候选，并从当前 automaton/plan node 或剩余任务重新调用规划器。
3. **Experiments**：如果已有机器人减少实验，Introduction 中只需一句；如果没有实验，最好不要在 Introduction 中强调，只作为 implementation capability。

11 可直接替换的 Introduction 逻辑版本

下面给出一个较完整的逻辑骨架。你可以把它拆分进现有 Introduction，而不是整段替换。

Skeleton for revised Introduction

1. Multi-robot systems are increasingly used in logistics, maintenance, inspection, and service scenarios. Many missions involve temporal ordering, synchronization, and collaborative actions, for which LTL provides a rigorous specification language.

2. In LTL-MRTA, the temporal formula specifies what collaborative propositions must be satisfied, but the concrete robots assigned to each proposition are not predetermined. This creates a large joint search space over automaton progress, robot choices, workspace travel, and synchronization times. Real deployments further introduce execution-side changes, including task updates and reductions in the available robot set.
3. Existing top-down methods include product-automaton search, MILP, and sampling-based synthesis. They are general but can be computationally expensive, especially before the first feasible solution is obtained.
4. Formula-structure-based methods, including decomposition, H-LTLf, and Poset-prod, reduce complexity by exploiting hierarchy, partial orders, or conjunctions of subformulas. These methods are effective when the mission naturally admits such a structure, but they require a suitable representation or decomposition and may be less direct for tightly coupled collaborative tasks.
5. Reactive methods handle task addition/removal or semantic-triggered replanning, but their main focus is updating the task logic or reachable automaton states after external events. They do not directly address the issue of injecting robot-team feasibility and intermediate planning information into the automaton-generation process for fast first-solution generation and subsequent optimization.
6. Therefore, this paper proposes a planning-aware automaton-generation framework. The key idea is to prune infeasible and unpromising automaton branches during GBA/NBA construction, build the NBA and planning graph synchronously, obtain an executable initial solution early, and then warm-start branch-and-bound to improve the final makespan. Robot availability reduction is handled by updating the available team and re-planning the remaining task as an execution mechanism.

12 参考依据说明

- 当前稿件已经明确使用 LTL2BA 的自动机构造管线,并在 GBA/NBA 构造中引入 planning-aware pruning、planning graph 和 warm-start BnB。因此, Introduction 的主线应围绕“automaton-generation-integrated planning”展开，而不是泛泛强调“fast”。
- Reactive MRTA 论文证明 dynamic task addition/removal 是合理的相关工作背景,其 automaton-reconstruction-based search 在检测到语义变化时重构任务公式和 Büchi automaton；它还将资源耗尽机器人视为 faulty 并从后续分配中排除，这可支持你轻量引入 robot availability reduction。
- Poset-prod 论文说明了子公式合取 + R-poset product 是处理长公式和在线任务释放的有效路线，但其出发点是 sc-LTL 子公式结构。你的差异化不应是否定它，而是强调你的方法不依赖用户预设层次/交集结构，而是在自动机构造过程中自动利用规划可行性和代价信息。
- H-LTLf 的定位是层次化 temporal-logic specification，提高复杂任务表达和搜索效率。你的方法应被写成 complementary：它解决方法层的构造和剪枝，而不是提出新的 specification

language。

13 最终建议

Introduction 修改后应呈现如下审稿人可接受的结论：

> 本文不是试图在 specification language 上取代 H-LTLf，也不是否定 Poset-prod 对长 sc-LTL 子公式合取的优势。本文的独特价值在于：当任务以全局 LTL-MRTA 形式给出、机器人分配未预先确定、协作约束存在耦合时，方法不要求用户先构造层次或子公式交集，而是在自动机生成过程中使用机器人团队和规划信息提前筛掉无效或低价值分支，从而更早得到可执行方案，并用该方案 warm-start 后续优化。机器人动态减少只作为执行中的 available robot set 更新来说明实践鲁棒性，不作为核心理论创新。